SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **CSA07**

Course Name: **Computer Networks**

Duration: **1: 30 Hr**

Date : 03.11.2026 Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Explain the layered network architectures (OSI/TCP-IP), the functions of each layer, and the fundamentals of data communication in computer networks.
CO2	Apply IP addressing, subnetting, and routing concepts to design and configure interconnected networks.

Answer All Questions

1. Answer the following questions on the OSI model and Transport layer services.

   (a) Explain the primary functions of the Data Link layer (framing, physical (MAC) addressing, medium access control, error detection using CRC, and hop-to-hopflow control) and of the Transport layer (process-to-process delivery using port numbers, segmentation and reassembly, end-to-end flow control, congestion control, and error recovery through acknowledgements and retransmission). Analyze how the two layers work together to give error-free and reliable data delivery: the Data Link layer detects and discards corrupted frames on each individual link, while the Transport layer provides end-to-end reliability across multiple links and routers by using sequence numbers, ACKs, and retransmission of lost segments. Use a layered diagram to support the explanation? *(6 Marks)*

   (b) Differentiate between connection-oriented service (TCP: handshake before transfer, reliable, ordered, flow and congestion control) and connectionless service (UDP: no setup, best-effort, low overhead) at the Transport layer. Compare them on connection setup, reliability, speed, header size, and state maintained. Give real-world examples: TCP for web browsing (HTTP/HTTPS), email (SMTP), and file transfer (FTP), and UDP for DNS queries, VoIP, video streaming, and online gaming. Present the comparison in a table. *(4 Marks)*

   (10) (CO1) (BL2) (BL4)

2. A data packet travels from a source host to a destination host across multiple routers.

   (a) Describe the concept of encapsulation, in which each OSI layer at the source adds its own header (and a trailer at the Data Link layer) to the data received from the layer above, producing the PDUs: data, segment or datagram, packet, frame, and bits. Describe decapsulation, in which each layer at the destination strips its corresponding header and passes the payload upward until the application receives the original data. Explain what happens at intermediate routers (the frame is decapsulated up to the Network layer, the next hop is determined, and a new frame is built). Support the answer with a neat diagram showing the headers at each layer. *(6 Marks)*

   (b) Identify the Network layer as the layer responsible for routing packets across interconnected networks. Explain how it uses logical addressing (IP addresses) to identify the source and destination networks and hosts, how routers use routing tables and the destination IP address to select the best path, and how logical (IP) addresses differ from physical (MAC) addresses, which change at each hop while IP addresses stay the same end to end. Mention fragmentation and the TTL field briefly. *(4 Marks)*

   (10) (CO1) (BL2)

(a) Routers and Layer 3 devices between a source and a destination system forward packets using IP addresses and routing tables. Explain the limitations of classful IP addressing: fixed network and host boundaries (Class A, B, C), wasted addresses (a Class C block of 254 hosts is too small and a Class B block of 65,534 is too large for most organizations), rapid IPv4 address exhaustion, inflexible subnetting, and the growth of routing tables on routers. Explain how Classless Inter-Domain Routing (CIDR) overcomes these problems through variable-length prefixes in slash notation (such as /22), allowing blocks of any suitable size, efficient address allocation, and route aggregation (supernetting) that reduces the size of routing tables and the load on routers. Give a short example of aggregating several contiguous networks into one prefix. *(5 Marks)*

(b) A customer site connects to a service provider's MPLS backbone through its Customer Edge (CE) router, which has the interface address 192.168.25.130 with the subnet mask 255.255.255.192 (/26). The provider edge (PE) router needs the customer's address block in order to associate it with a label-switched path (LSP) or VPN route. Determine, showing the binary working: (i) the Network ID (by ANDing the address with the mask), (ii) the Broadcast Address (by setting all host bits to 1), and (iii) the range of usable host IP addresses (first usable to last usable), along with the block size and the number of hosts in the block. *(5 Marks)*

(10) (CO1)(CO2) (BL2)

4. An organization is assigned the network address 172.16.0.0/16 and must divide it into at least 6 subnets for different departments while maximizing the number of usable hosts per subnet. Show the working at each step.

(a) Determine the minimum number of bits to borrow from the host portion (using $2^n \geq 6$) so that the usable hosts are maximized. Write the resulting subnet mask in both dotted-decimal and CIDR notation. *(3 Marks)*

(b) Calculate the total number of subnets created (2^n) and the number of usable host IP addresses per subnet ($2^h - 2$, where h is the number of remaininghost bits). Comment on the number of unused subnets left over. *(3 Marks)*

(c) For the first two subnets, list the Network ID, first usable IP, last usable IP, and Broadcast address. Show the block size (increment) used to find the second subnet, and present the results in a table. *(4 Marks)*

(10) (CO2) (BL3)

5. Answer the following questions on routing in network simulation tools.

(a) Compare and contrast static routing and dynamic routing in network simulation tools such as Cisco Packet Tracer. Compare them on configuration effort (manual versus automatic), adaptability to topology changes and link failures, convergence, scalability, router CPU and memory use, bandwidth overhead of routing updates, security, and administrative distance. Mention dynamic protocols such as RIP, OSPF, and EIGRP, and explain how a simulator lets you observe their behaviour. Discuss the conditions in which static routing is preferred, such as small networks, stub networks with a single exit path, default routes to an ISP, and situations needing predictability, low overhead, or control. Present the comparison in a table. (6 Marks)

(b) Outline the basic CLI configuration steps on a router to set up a static route directing traffic for the network 192.168.2.0/24 through the next-hop IPaddress 10.0.0.2. Include thesteps enable, configure terminal, ip route 192.168.2.0 255.255.255.0 10.0.0.2, exit, and saving the configuration with copy running-config startup-config. Then show how to verify the route with show ip route (looking for the S code) and test connectivity using ping. (4 Marks)

(10) (CO2) (BL3)(BL4)

CIA - 1

CSA0710 - Computer Networks

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1) a) Explain the data link & Transport layer:-

=> Data link layer:-

The Data link layer provides communication b/w directly connection devices it's main junction are framing: Divides data into frames

* MAC Address:- identifies devices on a local network
* Error Detection:- Detects error using CRC
* Flow control:- controls the speed of data transmission
* Access control:- manages access to the communication medium

=> Transport layer:-

The transport layer provides communication b/w application on different devices.

* Segmentation:- Divides data into smaller segments.
* Port Addressing:- Identifies applications
* Error Recovery:- prevents Receiver Overload.
* Reliable Delivery:- TCP provides ordered & Reliable data delivery.

b)

Connection - Oriented	connection less
Establishes a connection first TCP provides Reliable Delivery Ex:- File transfer using TCP	Does not Require connection setup UDP does not gurantee delivery Ex:- DNS Queries using UDP

2) Sol:

a) Encapsulation & capsulation:-

=> Encapsulation:- Each Osi layer add's it's header as do moves from sender to Receiver's network.

=> Decapsulation:- The Receiver Removes header layer by layer to obtain the Original data

=> Order :→ Segment → Packet → Frame → Bits

b) Network layer:-

* The Network layer performs routing b/w networks it uses logical ip addresses to identify destination and forward packets through routers.

3) Sol:-

a) classful Addressing & CIDR:-

* classful addressing uses fixed network size
* It wastes ip addresses and provide limited flexibility
* CIDR uses flexible prefix lengths such as 126
* It Reduces address wastage & supports route aggregation.

b) Subnet calculation:-

Given IP : 192.168.25.130

Mask :- 255.255.255.192 (126)

Network ID :- 192.168.25.128

Broadcast :- 192.168.25.191

First usable :- 192.168.25.129

Last usable :- 192.168.25.190

Usable hosts :- 62.

a) Subnets mask :-

Given : 172.16.0.0/16 at least 6 subnets

$2^3 = 8$ subnets

Borrowed bits :- 3

New CIDR :- 19

Subnet Mask :- 255.255.224.0

b) Subnet & Hosts :-

Total subnets :- $2^3 = 8$

Host bits : 32 - 19 = 13

usable hosts :- $2^{13} - 2$

$= 8190$

c) First two subnets :-

Details	Subnet 1	Subnet 2
Network ID	172.16.0.8	172.16.32.0
First usable	172.16.0.1	172.16.32.1
last usable	172.16.31.254	172.16.63.254
Broadcast	172.16.31.255	172.16.63.255

5) a) Static vs Dynamic Routing:-

=> Static Routing:- Routes are configured manually
Suitable for small, stable networks

=> Dynamic Routing:- Route are learned automatically
using protocols such as, RIP and OSPF

=> Static Routing uses fewer Resources while dynamic Routing adapts to network Changing.

b) Static Route configuration:-

Router > enable

Router # configure terminal

Router (config) # ip route 192.168.2.0
255.255.255.0 10.0.0.2

Router (config) # end

Router # show ip route: